

Sep 05, 2025

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

In the context of data protection and privacy, (a) give two possible definitions of privacy and briefly comment on them (are they overlapping? If not, how do they differ?), (b) explain the linkage attack on anonymized dataset with an example, and (c) explain whether using TLS protects client and server privacy by preventing a third party (who is not the client or server) from reading the transmitted messages. Additionally, indicate which definition of privacy you provided in your response to question (a) applies to this situation.

Question 2 [15 points]

Explain the framework of Attribute Based Access Control (ABAC), in particular how an access request is processed, and the reasons for which it has been introduced with respect to previous models (such as RBAC).

Question 3 [10 points]

Briefly describe what FIDO is and how it works.

Question 4 [10 points]

Briefly describe the difference between threats and vulnerabilities, providing an example for each one.

Part 2

Question 5 [25 points]

LLMs are a useful tool that can be used in many scenarios, but they need to be trained with the correct information. You have been hired by an AI-first company to assess whether their LLM provides the correct answers to cybersecurity topics.

You ask the LLM to describe what a **Cross-Site Scripting** is and how to mitigate it. It answers with:

Cross-Site Scripting (XSS) is a type of physical attack where a victim injects malicious SQL queries into a XSS to gain unauthorized access to a linked XSS. It allows the attacker to retrieve, modify, or delete sensitive data.

To mitigate XSS, developers should use strong passwords with salt and pepper, implement firewalls, and encrypt all data using TCP certificates. It is also helpful to scan the network regularly for vulnerabilities and block suspicious IP addresses.

Describe the errors and provide a more comprehensive version of the response.

Please, remember...

- on each used sheet of paper (possibly, 2 sheets at most), annotate:
 - Name
 - Surname
 - Student ID number
 - Question number
- write either in Italian or English, acronyms do not need to be translated;
- only pens are allowed. Drafts and text written using pencils will not be considered during the exam revision;
- adequately justify your answer. A phrase and a paragraph-long answer will have a significantly different grade;
- write clearly, your answers will not be *interpreted*;
- after turning in your exam, please respect your classmates by leaving quietly.